BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SUKKUR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2018

PHYSICS

Past Paper 2018 • Paper Code: A

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – C (MULTIPLE CHOICE QUESTIONS)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** A

Note: Q.1: Choose the correct answer from the given options and write the same in your answer copy.

Q1. The unit of thermodynamic temperature is:

(a) C° and F° (b) F° and K

(c) Only C° (d) Only K

Q2. Linear momentum and impulse has same:

(a) Dimension (b) Unit

(c) Physical significance (d) Both Dimension & Unit

Q3. If a vector $\vec{A}$ is multiplied by negative number ($n < 0$) then its direction is changed by:

(a) 0° (b) 90°

(c) 60° (d) 180°

Q4. The resultant of two forces of magnitude 5N each has also magnitude of 5N, the angle between the forces

(a) 0° (b) 90°

(c) 120° (d) 180°

Q5. Coplanar forces are those which are:

(a) Right angle to each other (b) Parallel to each other

(c) Anti parallel to each other (d) Acting in common plane

Q6. For uniform motion of a car along a straight line, the acceleration will be:

(a) Uniform (b) Variable

(c) Average (d) Zero

Q7. The range of projectile is directly proportional to:

(a) $\sin 2\theta$ (b) $\cos 2\theta$

(c) $\sin^2 \theta$ (d) $2 \sin \theta$

Q8. The unit of work in CGS system is:

(a) Joule (b) Newton

(c) Dyne (d) Ergs

Q9. Work energy principle is applied on:

(a) K.E (b) Gravitational P.E

(c) Elastic P.E (d) All of these

Q10. For positive angular displacement the rotation would be:

(a) Clockwise (b) Anti clockwise

(c) Parallel (d) Perpendicular

Q11. A man in an elevator descending with acceleration will conclude that his weight has:

(a) Increased (b) Decreased

(c) Remain same (d) Zero

Q12. When a bob of simple pendulum is at extreme position then it has maximum:

(a) K.E (b) P.E

(c) Both K.E & P.E (d) None of these

Q13. The component of weight mg sin θ in simple pendulum is:

(a) Along the centre (b) Perpendicular to string

(c) Along the string (d) None of these

Q14. Sonar is used to detect:

(a) Location of submarine (b) Depth of sea

(c) Control anti-submarine weapons (d) All of these

Q15. Beats are used to find:

(a) Frequency (b) wavelength

(c) Intensity (d) Speed

Q16. The electromagnetic wave theory was proposed by:

(a) Hertz (b) Maxwell

(c) Einstein (d) Huygen

Q17. The Collimator in a spectrometer is used to:

(a) Disperse the light beam
(b) Reflect the light beam
(c) Make the beam light parallel
(d) Converge the light beam

SECTION – A (Short Answer Questions)

Time Allowed: 2 Hours and 40 Minutes | **Maximum Marks:** 40

Note: ATTEMPT ANY EIGHT (08) QUESTIONS. ALL QUESTIONS CARRY EQUAL MARKS.

Q1. Define system of units and fundamental quantities. Give the units of seven base quantities.

Q2. Define Null vector. Find the unit vector parallel to the vector. $\vec{A} = 3\hat{i} + 6\hat{j} - 2\hat{k}$

Q3. State and explain Newton's 2nd law of motion and show that it leads to measurement of force.

Q4. What is projectile motion? Derive an equation for the height of projectile when its vertical velocity reduces to zero.

Q5. Describe a couple and moment of the couple.

Q6. What is the value of the gravitational acceleration at a distance of Thrice Earth's radius above the Earth's surface?

Q7. Define the term weightlessness. How is this inconvenience overcome in the space crafts?

Q8. What are the common defects of vision? How are they remedied?

Q9. What are Stationary waves and how are they formed? Distinguish between node and anti node.

Q10. State and explain the principle conservation of energy.

Q11. Explain briefly Newton's and Huygens's theory about nature of light.

SECTION – B (Long Answer Questions)

Time Allowed: 2 Hours and 40 Minutes | **Maximum Marks:** 28

Note: ATTEMPT ANY TWO (02) QUESTIONS. ALL QUESTIONS CARRY EQUAL MARKS.

Q12.

(a) [09] Describe Young experiment to demonstrate phenomenon of interference of light and hence derive an equation for fringe spacing.

(b) [05] A diffraction grating with 10000 lines per centimeter is illuminated by yellow light of wave length 589nm. At what angles Ist bright fringe is seen?

Q13.

(a) [09] What is simple harmonic motion? Show that acceleration of mass attached to spring is proportional to the displacement from mean position.

(b) [05] A block with a mass of 0.1Kg is attached to a spring of force constant 25N/m and is placed on a horizontal frictionless horizontal table. The spring is stretched by 20.cm. Compute the acceleration of block.

Q14.

(a) [09] Two bodies of different masses and having different velocities have an elastic collision in one dimension. Calculate their final velocities after collision.

(b) [05] Give any two special cases of elastic collision in one dimension.